C Programming Practice Set

Experiments 1-5 | EASY / FUNDAMENTALS

Q1. DATA TYPES, I/O & CASTING (Exp 1)

Write a C program to input a student's age (int), grade (char), and percentage (float).

a) Display all 3 values formatted neatly, ensuring percentage shows 2 decimal places.

b) Demonstrate explicit type casting by converting percentage to an integer and printing its whole-number value.

Q2. OPERATORS & INCREMENT TRACING (Exp 1)

Given: int x = 4, y = 3, z = 2, result;

a) Write a program to evaluate: result = x++ * ++y - z-- % 2 + x / y;

b) Print the final values of result, x, y, and z.

c) Show the step-by-step trace of how the result was calculated.

Q3. STANDARD STRING FUNCTIONS (Exp 5)

Write a C program that accepts two strings:

a) Print the length of both strings using strlen().

b) Compare them using strcmp() and display if they are equal or not.

c) Copy string 1 into a third array using strcpy(), join string 2 using strcat(), and print the final result.

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Experiments 1-5 | MEDIUM / INTERMEDIATE

Q4. DECISION MAKING & BRANCHING (Exp 2)

Write a C program to calculate an electricity bill based on units consumed:

- First 100 units: ₹2.00 / unit
- Next 200 units (101 to 300): ₹3.50 / unit
- Above 300 units: ₹5.00 / unit

* Add a 10% surcharge if the total bill exceeds ₹1000.

Q5. ITERATIVE LOGIC & NUMBERS (Exp 3)

Write a C program that accepts a positive integer N:

a) Reverse its digits using a loop (without using strings or arrays).

b) Check and print whether N is an Armstrong Number.

Q6. 1D ARRAY SEARCH & BOUNDS (Exp 4)

Write a C program to take N elements in a 1D array:

a) Find and print the Maximum and Minimum elements along with their index positions.

b) Take a search key from the user, perform a Linear Search, and print all indices where it appears.

Q7. STRING INPUT & CHARACTER COUNTING (Exp 5)

Write a C program that reads a full sentence using fgets():

a) Count and print total vowels, consonants, spaces, and total characters.

b) Explain why using scanf("%s", ...) is bad for reading sentences.

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Experiments 1-5 | HARD / ADVANCED

Q8. MATRIX OPERATIONS & PATTERNS (Exp 3 & 4)

Write a C program to:

a) Read a 3x3 matrix, display its row sums, column sums, and its Transpose matrix.

b) Print an inverted pyramid star pattern for N rows:

```
* * * * *
  * * *
    *
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Q9. ARRAY SORTING & IN-PLACE ROTATION (Exp 4)

Write a C program for an array of size N:

a) Sort the array in ascending order using Bubble Sort.

b) Rotate the sorted array to the left by K positions IN-PLACE (without using a 2nd array).

Q10. CUSTOM STRING PROCESSING & PALINDROME (Exp 5)

Write a C program to read a string using fgets():

a) Remove all spaces from the sentence in-place.

b) Convert uppercase characters to lowercase manually without using tolower().

c) Check and display whether the final string is a Palindrome.